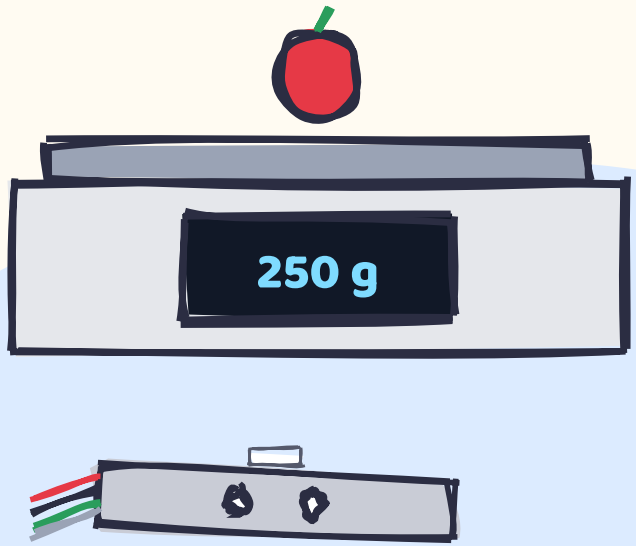
Junior Makers · Class 26

Smart Kitchen Scale

Weigh anything to the gram, and make it smart

Load cell | Calibration | OLED | Target alarm



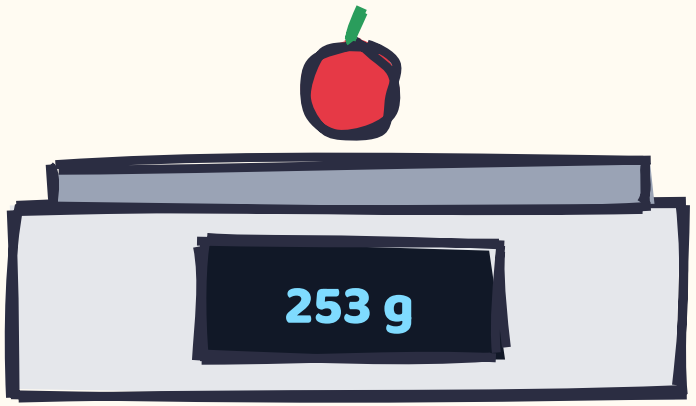
Did you know?
Digital scales in shops must be checked by the government so customers always get the weight they pay for.

Let's think...

How does a scale know the weight?

No springs?

Old scales had a big spring that stretched, and a needle that moved. Digital scales have no needle and almost nothing moves. Yet they say **253 g** instantly. What is hiding inside?



A tiny spring?

Magic chip?

A bending metal bar!

Today: you build a **digital kitchen scale** with a load cell, then make it smart with a **target-weight alarm**.

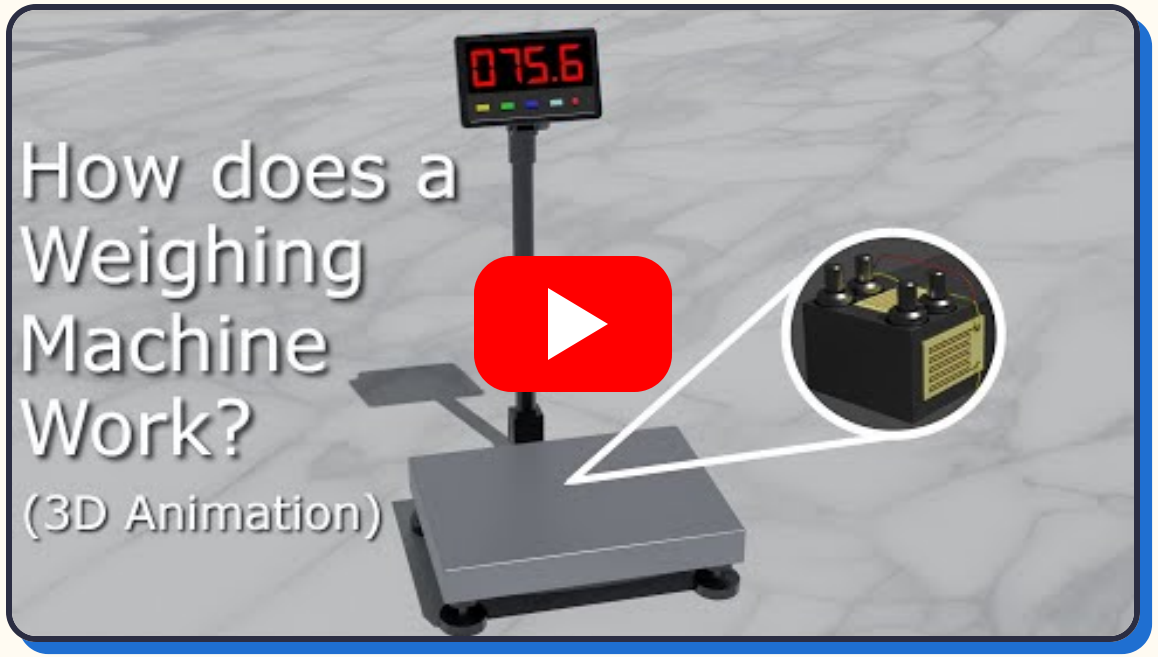
Learning objectives

By the end of this class I can...

1	**Load cell:** explain how a **load cell** turns a tiny bend into a number	**Proof:** explain 'strain' with the ruler example
2	**Calibrate:** **tare** (zero) and **calibrate** my scale with a known weight	**Proof:** my scale reads a 100 g water cup within 5 g
3	**Display:** show the weight in grams on an **OLED**, with a tare button	**Proof:** pressing the button sets it back to 0 g
4	**Smart feature:** add a **target-weight alarm** for recipes or portions	**Proof:** the buzzer beeps when I reach the target

Objective 1 · Load cell · video

Video time!



How does a Weighing Machine Work?

3D Requiem · about 4 minutes · youtu.be/lZafPys6x5M

Watch for...

1. What part of the scale **bends** when you put something on it?
2. How does a tiny bend become a **number**?
3. Why must a scale be set to **zero** first?

Scan to watch

After the video: Press gently on a ruler on the desk. Can you see it bend? A load cell bends far less than that!

Objective 1 · Load cell

A load cell is a bar that bends a tiny bit

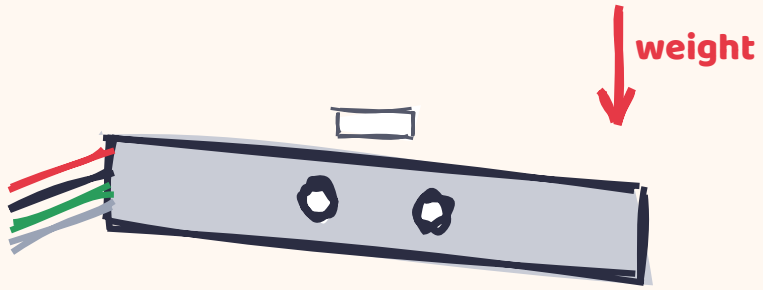
The idea

The load cell is an aluminium bar with a **strain gauge** glued on: a thin zig-zag wire. When weight bends the bar (much less than a hair's width!), the wire stretches and its **resistance changes** very slightly.

The change is so tiny that we need an **HX711** chip to amplify it and turn it into a number.

Gentle!
Never press on the load cell with your fingers or drop things on it: too much force bends it for ever.

Example: a ruler on a desk



Hang a ruler off the desk and press the end: it bends. Press harder: it bends more. The load cell bends the same way, just far, far less, and a sensor measures exactly how much.

Objective 2 · Calibrate

Tare and calibrate

The idea

Tare: with nothing on the plate, tell the scale 'this is 0 g'. (Bakers tare to ignore the bowl!)

Calibrate: put on something you KNOW the weight of. Water is perfect: **100 ml of water = 100 g**.

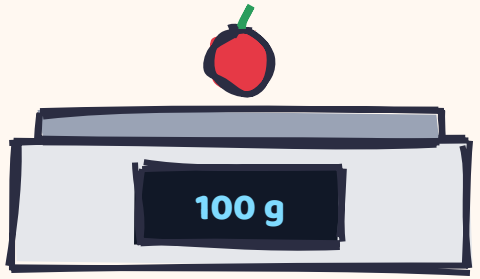
scale factor = raw change ÷ known grams

e.g. raw went up 21,400 for 100 g → factor = **214**

Tip

Tare the empty cup first, then pour exactly 100 ml using a measuring jug or syringe.

Example: a new ruler with no numbers



100 ml water = 100 g

Imagine a ruler with lines but no numbers. You measure a 10 cm card and write '10' at that line. Now the ruler is useful! Calibration writes the numbers on our scale.

Objective 2 · Calibrate

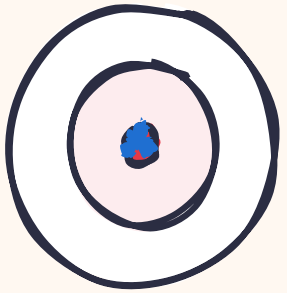
Accurate and precise

The idea

Accurate: the reading is close to the true weight (100 g reads 100 g).
Precise: the reading is the same every time (100, 100, 101, 100).

Test it: weigh the same object **5 times**. Do the numbers jump around? Take the **average** of 10 readings to smooth out the wobble.

Example: darts on a board



both!

Darts all bunched in a corner: precise, not accurate. Darts spread around the bullseye: accurate on average, not precise. All in the bullseye: both! Good scales are both.

Objective 4 · Smart feature

Make it smart: a target alarm

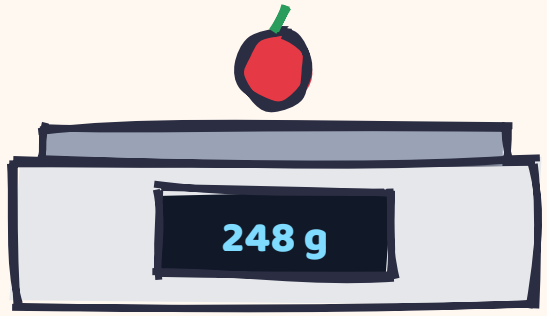
The idea

A **target weight** turns a scale into a helper:

- Recipe mode: 'pour flour until **250 g**'
- The LED turns on at **90%** of the target (slow down!)
- The buzzer beeps at the target

Use **if** blocks to compare the weight with the target.

Example: filling a glass



target 250 g

When you pour water into a glass, you slow down as it gets near the top. Our LED is the 'slow down' warning and the buzzer is the 'stop!'.

Real world

Load cells everywhere



Shops
Fruit and meat scales that print a price



Airports
Check-in belts weigh your luggage



Weighbridges
Weigh whole trucks on the motorway



Hospitals
Baby scales and beds that weigh patients

Why must a shop's scale be checked by inspectors?

Fairness
If a scale reads 10% heavy, every customer pays 10% too much. Calibration is about honesty.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Guess the weight
2. Our scale kit (per group)
3. Build and wire the scale
4. See the raw numbers
5. Grams on the screen
6. Recipe alarm
7. Guess the weight: results!

Class game

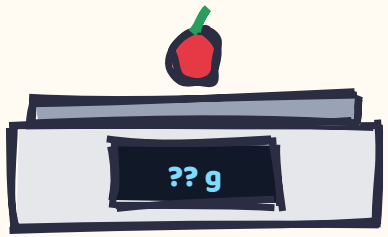
Guess the weight

How to play

1. Pass around 5 mystery objects: a pencil case, an apple, a water bottle, a book, a shoe.
2. Each team writes its **guess** in grams for each one.
3. Keep the guesses secret until our scale is built.
4. At the end of class, weigh them all. Which team was closest overall?

Then discuss

Were your guesses too high or too low? Why are humans bad at guessing weight?



Big idea

We need instruments because our senses are not precise. That's why we build and calibrate scales.

Build it

Our scale kit (per group)



Arduino Uno
The scale's brain



5 kg load cell + HX711
Senses and amplifies the bend



OLED display
Shows the grams



Push button + 10k
The TARE (zero) button



Buzzer + LED
Target alarm



Cardboard
Base and weighing plate

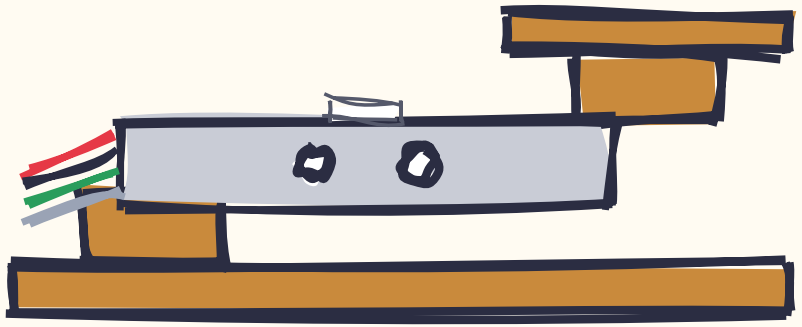
The load cell must be fixed at ONE end to the base and hold the plate at the OTHER end, so it can bend. Use the screw holes.

Max 5 kg
Never put more than 5 kg on it. A heavy textbook pile could ruin it.

Objective 1 · Load cell · Build step 1 of 4

Build and wire the scale

Part	Connect to
Load cell red / black	HX711 E+ / E–
Load cell white / green	HX711 A– / A+
HX711 VCC / GND	5V / GND
HX711 DT / SCK	pin 3 / pin 2
OLED SDA / SCL	A4 / A5
Button (10k to GND)	pin 4 and 5V



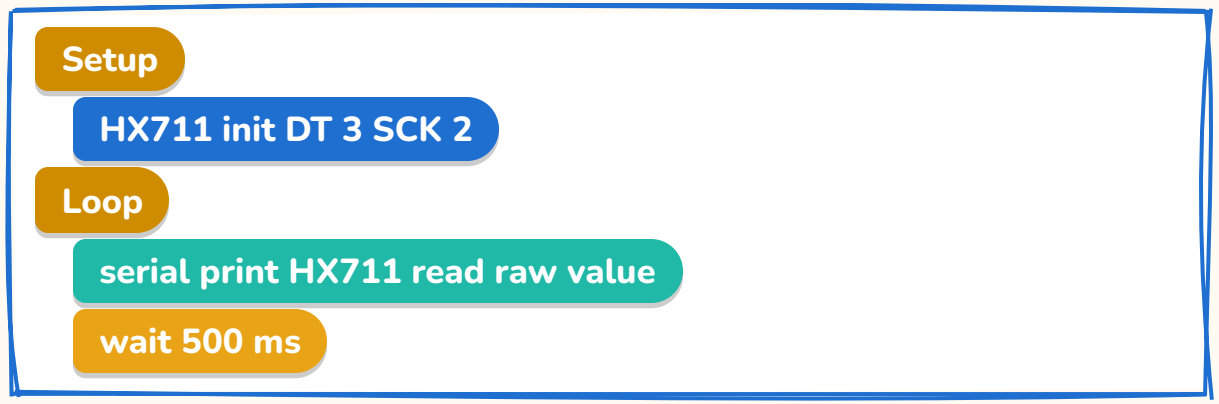
fixed end · free end

The Z shape

Base → spacer → load cell → spacer → plate. The gaps let the bar bend.

Objective 1 · Load cell · Build step 2 of 4

See the raw numbers



Test	Raw number
Empty plate	
Press very gently	
100 g of water in a cup	

Big, strange numbers

You'll see numbers like 84,512 or −12,300. They change when you add weight, but they are NOT grams yet. We need to **calibrate**.

Wrong way?
If the number goes DOWN when you add weight, swap the white and green wires (or turn the cell over).

Objective 3 · Display · Build step 3 of 4

Grams on the screen

```
Setup
  HX711 init DT 3 SCK 2
  HX711 set scale 214 (YOUR factor)
  HX711 tare
Loop
  set grams to round( HX711 get weight )
  if button pin 4 pressed then HX711 tare
  OLED clear · text size 3 · print grams · print ' g' · show
  wait 200 ms
```

Test	Should be	We got
100 ml water	100 g	
200 ml water	200 g	
Empty, after tare	0 g	

Not quite?
Reads 95 g for 100 g? New factor = 214 × 95 ÷ 100.
Try again!

MAKE

Objective 4 · Smart feature · Build step 4 of 4

Recipe alarm

Setup: set target to 250

Loop

- (read grams and show on OLED as before)
- if grams ≥ target × 0.9 then LED pin 12 ON else LED OFF
- if grams ≥ target and not beeped then
  - tone pin 11 1000 Hz 300 ms · set beeped to true
- if grams < 5 then set beeped to false

Why 'beeped'?

Without it, the buzzer would beep forever while the bowl sits on the scale. It resets when the plate is empty.

Change target

Use the button: short press = tare, long press = add 50 g to the target.

Objective 4 · Smart feature · test it

Guess the weight: results!

Object	Our guess	Scale says	Difference
Pencil case			
Apple			
Water bottle			
Book			
Shoe			

Total difference

Add up all the differences (ignore minus signs). Lowest total wins!

Test like an engineer

Bug hunt

Bug	Likely cause	Did it happen?
Reading always 0 or stuck	DT/SCK swapped or loose wire	
Number goes down with weight	A+/A− swapped	
Reading slowly drifts	load cell warming up / bumping	
Wrong grams	scale factor not calibrated	
Very jumpy numbers	plate touching the base	

Debug checklist

Check the Z-shape: the plate must ONLY touch the load cell, nothing else.

Let it sit for 1 minute after switching on, then tare. Electronics like to warm up.

Level up!

Scale challenges

Choose your challenge



Price scale: type a price per kg; show the cost of what's on the plate.



Food waste: weigh lunch leftovers each day and add up the week.



Counting scale: weigh 10 coins, then COUNT a pile of coins by weight!



Units: a button that switches between grams and ounces.

Think like an engineer

If 10 identical coins weigh 52 g, how many coins are in a pile that weighs 260 g?

Boss level

Make the scale average 10 readings for a steadier display.

Objective check

Can I really do it? Prove it.

01	What bends inside the scale, and what does the HX711 do?	○ ○ ○
02	Explain tare and calibrate. What is YOUR scale factor?	○ ○ ○
03	Weigh 100 ml of water. How close is your scale?	○ ○ ○
04	Show your recipe alarm reaching the target.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Scale accuracy contest: weigh the teacher's mystery object. Closest scale wins.

Show what you know

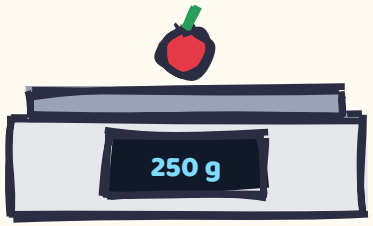
What did we learn?

- [x] A load cell is a bar that bends a tiny bit under weight
- [x] The HX711 amplifies the tiny change into a number
- [x] Tare = set zero; calibrate = match a known weight
- [x] Accurate = close to the truth; precise = repeatable
- [x] if blocks turn a scale into a smart helper

Exit ticket: your scale reads 90 g for 100 ml of water. What do you change?

At home

Weigh 3 things at home on a kitchen scale. Bring the numbers to test our scales!



Next class: Arcade Claw Machine!